Pouya Parsa

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Research Interests: Vision-Language Models • Memory-Augmented Large Language Models

Education

Master of Science — **Minneapolis, USA**
University of Minnesota Twin Cities — *2024–2026*

- GPA: 3.7/4.00; selected coursework: Computer Vision and Generative AI.

Bachelor of Science in Computer Science — **Tehran, Iran**
Amirkabir University of Technology (Tehran Polytechnic) — *2018–2022*

- GPA: 3.72/4.00; thesis: *Transformer-Based Deep Learning Method for Portfolio Optimization.*

Publications

Visual Distribution Anchoring for Efficient Prompt Tuning
Pouya Parsa, Raoof Zare Moayedi, and Seongjin Choi — *arXiv:2607.28967, July 2026*

- Introduced class-specific visual prototypes from an unlabeled target pool, improving mean frozen vision–language classification accuracy from 65.82% to 69.21% without target-side optimization or test-query access.

Can the Cloud Drive? Infrastructure Feasibility of Offloading Autonomous Driving Across 5G and 6G
Pouya Parsa, Kawon Han, and Seongjin Choi — *arXiv:2607.09045, 2026*

- Analyzed cloud offloading for vision–language–action driving models, showing when GPU memory bandwidth remains the bottleneck after 5G/6G uplink constraints are satisfied.

Video-based Vehicle Surveillance in the Wild: License Plate, Make, and Model Recognition with Self Reflective Vision-Language Models
Pouya Parsa, Keya Li, Kara M. Kockelman, and Seongjin Choi — *arXiv:2508.01387; TRB 2026*

- Developed a video vision–language pipeline achieving 91.67% license-plate and 66.67% make-and-model top-1 accuracy on smartphone data; self-reflection improved make-and-model results by 5.72% on average.

Where2Start: Leveraging Initial States for Robust and Sample-Efficient Reinforcement Learning
Pouya Parsa, Raoof Zare Moayedi, Mohammad Bornosi, and Mohammad Mahdi Bejani — *arXiv:2311.15089, 2023*

- Proposed an informative initial-state selection method that achieved up to an 8× improvement in sample efficiency over reinforcement-learning baselines.

Experience

Research Assistant — **Minneapolis, USA**
University of Minnesota; advisor: Dr. Seongjin Choi — *Aug 2024–Present*

- Conduct research in machine learning with an emphasis on vision–language models, computer vision, and generative AI.
- Develop methods for self-reflective recognition, efficient prompt tuning, and infrastructure-aware autonomous-driving inference.

Project Manager — **Remote (Dubai)**
XTON — *Apr 2024–Jul 2024*

- Managed infrastructure for a high-availability crypto clicker application serving more than 500,000 users.
- Implemented a serverless architecture using AWS Lambda and auto-scaling and coordinated stakeholder expectations.

Machine Learning Engineer — **Remote (Germany)**
ZebraCat.ai — *Sep 2022–Mar 2023*

- Developed embedding-based video retrieval and semantic video-segmentation methods.
- Collected a dataset of more than one million advertisement videos.

Data Science Intern — **Tehran, Iran**
MCI (Hamrahe Aval) — *Jul 2022–Sep 2022*

- Built a fault-tolerant ETL pipeline and collaborative-filtering service for recommender systems at Iran's largest mobile operator, delivering recommendations in under 10 ms.

Machine Learning Engineer Intern — **Zurich, Switzerland**
University of Zurich — *Jan 2022–Mar 2022*

- Developed interfaces for data ingestion, preprocessing, machine learning, visualization, and report generation.

Honors and Awards

Departmental Fellowship Award — **2026**
University of Minnesota Twin Cities
Awarded in recognition of academic excellence and research potential.

CTS TRB Reimbursement Award — **2026**
Center for Transportation Studies
Received competitive travel reimbursement to present research at TRB 2026.

Outstanding Student Recognition
Amirkabir University of Technology Honors and Olympiads Program
Recognized for academic excellence; ranked 403rd among approximately 150,000 participants in Iran's national university entrance examination.

Technical Skills

Programming: Python, Java, C/C++
Machine Learning: PyTorch, TensorFlow, scikit-learn, OpenCV
Data: NumPy, pandas, SQL, NoSQL

Selected Projects

Heart Rate Measurement from Video — Project link
Estimated heart rate non-invasively from subtle color changes in facial video.

Neural-Network Optimizer Comparison — Project link
Compared particle swarm optimization with Adam for neural-network training.

Movie Recommender — Project link
Implemented collaborative filtering with singular value decomposition.

Teaching and Service

Teaching Assistant: Computer Vision (Winter 2023); Optimization in Neural Networks (Winter 2023); Artificial Intelligence (Winter 2022).
Service: Board Member, Scientific Association, Amirkabir University of Technology (2020–2021); Member, Khayam–Turing Machine Learning and Data Science Competition Group (2020–2021).

Languages

English: Fluent (TOEFL iBT 104/120) **Persian:** Native